

Song Lab

Hear chords, build beats, and map your song. Scan the code or type:

cmfnyc.github.io/song-lab/



Song Anatomy Map

Every section has a job

- **Intro** — Sets the mood. Says "a song is starting." Often just chords or a riff.
- **Verse** — Tells the story. Same melody every time, **new words** every time.
- **Pre-chorus** (optional) — Builds up. Makes you feel "something big is coming."
- **Chorus** — The main idea and usually the **title**. Same melody **and** same words every time. The biggest part.
- **Post-chorus** (optional) — A short, catchy add-on after the chorus. Often just "oh-oh" or one word.
- **Bridge** — Something **different**, usually once, near the end. New chords, new angle.
- **Outro** — The ending. Fades out, or stops on a big hit.

The hook is the part that gets stuck in your head. Usually a short phrase in the chorus. Sometimes it's an instrument riff.

The most common pop shape

Verse → Chorus → Verse → Chorus → Bridge → Chorus

With extras: Intro → Verse → Pre-chorus → Chorus → Verse → Pre-chorus → Chorus → Bridge → Chorus → Outro

What changes when the chorus arrives?

Listen and check every one you hear:

- ☐ It gets louder
- ☐ The melody goes higher

- ☐ More instruments come in
- ☐ The drums change
- ☐ The title is sung
- ☐ The words repeat
- ☐ Other: _____

These are the tools **you** will use in your own song.

Listening Map

Song: _____ **Artist:** _____

Tempo guess: slow / medium / fast **Mood in one word:** _____

Write each section as it happens.

1. Time _____ Section _____ What I notice _____
2. Time _____ Section _____ What I notice _____
3. Time _____ Section _____ What I notice _____
4. Time _____ Section _____ What I notice _____
5. Time _____ Section _____ What I notice _____
6. Time _____ Section _____ What I notice _____
7. Time _____ Section _____ What I notice _____
8. Time _____ Section _____ What I notice _____

The hook (5 words or fewer): _____

How many times does the hook repeat in one chorus? _____

One thing I want to steal for my song (a structure, a sound, a trick — not the melody or words):

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Chord Cheatsheet

Roman numerals in 30 seconds

Chords get numbers based on their position in the key. The numbers let you move any pattern into any key.

- **UPPERCASE** = major chord (bright)
- **lowercase** = minor chord (darker)

Number	I	ii	iii	IV	V	vi
Key of C (home key)	C	Dm	Em	F	G	Am
Key of G (optional)	G	Am	Bm	C	D	Em

The **I** chord is **home**. Songs feel finished when they land there.

(There is also a **vii°** chord — Bdim in C — but pop songs rarely use it. Skip it.)

Ukulele shapes

Four numbers = the four strings, top to bottom: **G C E A**. 0 = open string.

Key of C

- C = 0003
- Dm = 2210
- Em = 0432
- F = 2010
- G = 0232
- Am = 2000

Key of G

- G = 0232
- Am = 2000
- Bm = 4222 (harder)
- C = 0003
- D = 2220
- Em = 0432

Strum patterns, melody notes, and Phase 2 chords are in [Ukulele Songwriter's Guide](#).

What's in each chord (letter names)

You need these for the chord chart beside おんぷノート, and for playing chords on a GarageBand keyboard. A chord is three notes: skip a letter, take one, skip, take one.

Key of C

- C = C E G
- Dm = D F A
- Em = E G B
- F = F A C
- G = G B D
- Am = A C E

Key of G

- G = G B D
- Am = A C E
- Bm = B D F#
- C = C E G
- D = D F# A
- Em = E G B

On a GarageBand keyboard: right hand plays the three notes, left hand plays just the chord's letter name, low.

Famous four-chord loops

Try each one. Write a mood word next to it.

1. **I - V - vi - IV** — C G Am F / G D Em C — Mood: _____
2. **vi - IV - I - V** — Am F C G / Em C G D — Mood: _____
3. **I - vi - IV - V** — C Am F G / G Em C D — Mood: _____
4. **IV - I - V - vi** — F C G Am / C G D Em — Mood: _____
5. **I - IV - vi - V** — C F Am G / G C Em D — Mood: _____

Make your own: pick any four chords from I, ii, iii, IV, V, vi. Rules of thumb:

- Starting on **I** feels settled and bright.
- Starting on **vi** feels emotional.
- Ending on **V** makes you want to go back to the start. Great for loops.

My Song 1 progression: _____ - _____ - _____ - _____ **Tempo:** _____ BPM

Phase 2 chords

See [Minor & Borrowed Chords](#) for minor keys and borrowed chords (Fm, B^b, A^b in C).

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Beat Patterns

How to read the grid

One bar of 4/4 is split into 16 steps. Count it out loud:

1 e & a 2 e & a 3 e & a 4 e & a

- The numbers (1, 2, 3, 4) are the **beats** — where you tap your foot.
- The "&"s are halfway between beats.
- The "e" and "a" are the in-between-in-between (sixteenth notes).

X = hit. **.** = rest.

The pop drum kit

- **Kick** — low "boom." Gives the song weight.
- **Snare / Clap** — sharp "crack" on beats 2 and 4. The part you clap along to.
- **Hi-hat** — "tss." Keeps time and adds energy.

Golden rule: keep the snare on 2 and 4. Change anything else.

1. Basic Rock / Pop

Works for almost everything.

	1	e	&	a	2	e	&	a	3	e	&	a	4	e	&	a
Hi-hat	X	.	X	.	X	.	X	.	X	.	X	.	X	.	X	.
Snare	.	.	.	.	X	.	.	.	.	.	.	.	X	.	.	.
Kick	X	.	.	.	.	.	.	.	X	.	X	.	.	.	.	.

2. Four on the Floor

Dance, disco, EDM. Kick on every beat.

	1	e	&	a	2	e	&	a	3	e	&	a	4	e	&	a
Hi-hat	.	.	X	.	.	.	X	.	.	.	X	.	.	.	X	.
Clap	.	.	.	.	X	.	.	.	.	.	.	.	X	.	.	.
Kick	X	.	.	.	X	.	.	.	X	.	.	.	X	.	.	.

3. Half-Time

Feels slower and heavier, even at the same tempo. Snare only on beat 3.

	1	e	&	a	2	e	&	a	3	e	&	a	4	e	&	a
Hi-hat	X	.	X	.	X	.	X	.	X	.	X	.	X	.	X	.
Snare	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.
Kick	X	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.

4. Ballad

Slow and spacious. Hi-hat on beats only.

	1	e	&	a	2	e	&	a	3	e	&	a	4	e	&	a
Hi-hat	X	.	.	.	X	.	.	.	X	.	.	.	X	.	.	.
Snare	.	.	.	.	X	.	.	.	.	.	.	.	X	.	.	.
Kick	X	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.

5. Trap

Hip-hop and modern pop. Fast hi-hats, half-time snare.

	1	e	&	a	2	e	&	a	3	e	&	a	4	e	&	a
Hi-hat	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Snare	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.
Kick	X	.	.	.	.	.	.	.	X	.	.	X	.	.	.	.

6. Dembow (Reggaeton)

Latin pop. The snare pattern is the signature.

	1	e	&	a	2	e	&	a	3	e	&	a	4	e	&	a
Snare	.	.	.	X	.	.	X	.	.	.	.	X	.	.	X	.
Kick	X	.	.	.	X	.	.	.	X	.	.	.	X	.	.	.

Make the chords move too

Play your progression three ways. Which fits your song?

- **Whole notes:** one hit per bar, let it ring. → ballad
- **Quarter notes:** hit on 1, 2, 3, 4. → driving
- **The push:** hit on 1, then on the "&" of 2, then hold. Say it: **ONE** two **AND** three four. → modern pop

On ukulele, these become strum patterns, plus two more (island strum and the chuck). See [Ukulele Songwriter's Guide](#).

My beat: _____ **My strum:** _____

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Melody & Hook Toolkit

The Hook Recipe

1. **Find a rhythm you can tap.** Tap it on the table over your loop.
2. **Put it on 2–3 notes.** That's all you need.
3. **Repeat it.** Two, three, or four times.
4. **Change the ending** the last time. Go higher, or land on the home note.

Many famous hooks use only two or three different notes. The **rhythm** is what makes them catchy.

Six rules for a melody people remember

1. **Repeat.** Repetition is catchiness.
 2. **Keep phrases short.** 2–5 words, then a breath.
 3. **Mostly steps, one leap.** Move to neighbor notes most of the time. Save one big jump for the most important word.
 4. **Chord tones on strong beats.** On beats 1 and 3, use a note that's in the chord. It will sound "right." (Chord notes are on the Chord Cheatsheet.)
 5. **Chorus higher than verse.** Higher = more energy.
 6. **Leave space.** Rests let the listener breathe and remember.
-

Catching ideas

- Loop your groove and **hum** over it. Use "la," "na," "oh." Words come later.
- Record **everything** on Voice Memos. Ideas vanish in minutes.
- Do **three takes** without stopping to judge. Then listen and pick.

- No singing? Find the melody on your instrument, one note at a time.
-

Writing it down (おんぷノート)

1. Make a **chord chart** on paper: bar numbers, the chord for each bar, and its three notes. (The app can't show chord names.)
2. In the app, set the **key signature and 4/4**.
3. **Handwrite** notes one bar at a time. Zoom in; a stylus helps. **Play back after every bar**.
4. Check beat 1 of each bar: is the note on your chord chart for that bar?
5. Sour note? Move it **one step up or down**.
6. Try different **playback sounds**. Which one sounds like your song?
7. **Export a PDF** before you finish. The app keeps only one score.

More tips: [Using おんぷノート](#)

The One-Listen Test

Play your chorus for someone once. Can they hum the hook back?

- **Yes** → you have a hook.
 - **No** → simplify the rhythm, repeat more, or use fewer notes.
-

Level 2 tricks

- **Question and answer:** first phrase ends "up in the air" (unfinished); second phrase ends on the home note (finished).
 - **Pickup:** start the melody just before beat 1.
 - **Anticipation:** land an important note half a beat early (on the "&" before the beat). Sounds modern and groovy.
 - **Displacement:** repeat the hook but start it one beat later.
 - **Air pocket:** a moment of silence right before the chorus.
-

My hook

Hook words (or syllables): _____

Rhythm (tap it, then write dashes for each note): _____

Notes it uses: _____

Passed the One-Listen Test? yes / not yet

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Lyric Toolkit

Start with the title

The chorus exists to deliver the title. Good pop titles are:

- **Short** (1–4 words)
- **Concrete** (something you can picture or feel)
- **Good to sing** (open vowels like "oh," "ay," "ee" hold well on long notes)

Title brainstorm (write 10, fast, no judging):

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____

Chorus vs. verse

Chorus = the big idea. Simple. Repeats. Anyone can sing along.

Verse = the details. Who? Where? What happened? Specific things you can see and hear.

Show, don't tell

- Telling: "I'm nervous."
- Showing: "My hands won't stop shaking on the mic."
- Telling: "I miss my friend."

- Showing: "Your number's still the first one on my phone."

Verses should **show**.

Object writing (10 minutes)

Pick an object, place, or moment. Write for 10 minutes without stopping. No rhyming. No fixing. Use all seven senses:

1. **See** — colors, shapes, light
2. **Hear** — sounds, voices, silence
3. **Smell**
4. **Taste**
5. **Touch** — textures, temperature
6. **Body** — heartbeat, tired, tense, relaxed
7. **Movement** — what's moving, how fast

When the timer ends, circle your 3–5 best phrases. Use them in your verses.

(Object writing comes from Pat Pattison of Berklee College of Music, author of "Writing Better Lyrics.")

Rhyme

Perfect rhyme: night / light, go / slow. Clean, but can sound predictable.

Near rhyme: night / mine, home / alone, heart / hard. Used all the time in modern pop. Sounds more natural.

Rhyme schemes (letters show which lines rhyme):

- **AABB** — lines 1+2 rhyme, lines 3+4 rhyme
- **ABAB** — lines 1+3 rhyme, lines 2+4 rhyme
- **XAXA** — only lines 2+4 rhyme (easiest, very common)

Internal rhyme (Level 2): a rhyme inside one line. "I *fake* a smile, I *take* the long way home."

Fitting words to the melody

1. Under each note of your melody, draw a dash: —. One dash = one syllable.
 2. Fill in the dashes with words.
 3. **Check the stress.** Every word has a loud syllable: to-**NIGHT**, re-**MEM**-ber, **SUM**-mer.
The loud syllable should land on a strong beat.
 4. If a word sounds wrong when sung, **move the word.** Don't force it.
 5. Too many syllables? Cut one. "Going to" → "gonna" is fine in pop.
-

Stuck? Topic menu

A place you know really well • the night before something big • a game you love • someone who moved away • a superpower • being misunderstood • a pet's point of view • a secret • the first day of summer • a message you never sent • a silly song about something ordinary

Lyric checklist

- ☐ The title is in the chorus (probably more than once)
- ☐ The chorus is simple enough to sing along to
- ☐ Each verse has at least one thing you can see or hear
- ☐ No twisted word order just to rhyme
- ☐ Stressed syllables land on strong beats
- ☐ Verse 2 moves the story forward

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Song Map Planner

How long will my song be?

Seconds = bars × 4 × 60 ÷ BPM

Example: 40 bars at 100 BPM → $40 \times 4 \times 60 \div 100 = 96$ seconds (1:36).

Song 1 map (First Song form)

Title: _____ **Key:** _____ **Tempo:** _____ BPM

1. **Intro** — 4 bars — What plays: _____ Energy (1–5): ____
2. **Verse 1** — 8 bars — What plays: _____ Energy: ____
3. **Chorus** — 8 bars — What plays: _____ Energy: ____
4. **Verse 2** — 8 bars — What plays: _____ Energy: ____
5. **Chorus** — 8 bars — What plays: _____ Energy: ____
6. **Outro** — 4 bars — What plays: _____ Energy: ____

Total bars: _____ **Length:** _____

Song 2 map (Full Pop form)

Title: _____ **Key:** _____ **Tempo:** _____ BPM

1. **Intro** — 4 bars — _____ Energy: ____
2. **Verse 1** — 8 bars — _____ Energy: ____
3. **Pre-chorus** — 4 bars — _____ Energy: ____
4. **Chorus** — 8 bars — _____ Energy: ____
5. **Verse 2** — 8 bars — _____ Energy: ____

6. **Pre-chorus** — 4 bars — _____ Energy: ____
7. **Chorus** — 8 bars — _____ Energy: ____
8. **Bridge** — 8 bars — _____ Energy: ____
9. **Final Chorus** — 8-16 bars — _____ Energy: ____
10. **Outro** — 4 bars — _____ Energy: ____

Lift for the final chorus: key change / new layer / drop and return / ad-libs

Energy curve

Draw a line across this page from left to right. Up = bigger, down = smaller.

A typical pop curve:

- Intro: low
 - Verse 1: medium
 - Chorus: high
 - Verse 2: medium, a bit higher than verse 1
 - Chorus: high
 - Bridge: drop down *or* build up
 - Final chorus: **highest**
 - Outro: low
-

Ways to raise energy

- Add an instrument
- Drums get busier (more hi-hat, more kick)
- Melody goes higher
- Add a harmony or double the vocal
- A fill or riser right before the section

Ways to lower energy

- Mute instruments (the mute button is your best tool)

- Simpler drums, or no drums
- Lower melody, fewer words
- Chords held longer

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Mix & Export Checklist

Before you record

- ☐ Quiet room, soft surfaces (a closet full of clothes works great)
- ☐ Headphones on, so the music doesn't leak into the mic
- ☐ Mouth about one hand-width from the mic
- ☐ Warm up: hum, lip trills, sing the chorus once quietly
- ☐ Save a new version first (-v2 , -v3 ...)

Recording

- ☐ Record the chorus first (you know it best)
 - ☐ 3 takes per section, no stopping to judge
 - ☐ Listen back and keep the best take (or best phrases)
 - ☐ Optional: sing the chorus a second time on a new track (a "double") and pan the two slightly apart
-

The four mix moves

1. Balance

- ☐ The **lead** (vocal or melody) is the loudest thing
- ☐ Drums next
- ☐ Everything else sits just under the lead
- ☐ Trick: turn everything down, then bring the lead up first, then add the others one at a time

2. Pan

- ☐ Center: kick, snare, bass, lead vocal
- ☐ Slightly left: chords (or one guitar)
- ☐ Slightly right: extra instrument (or second guitar)

3. Reverb

- ☐ A little reverb on the vocal so it sounds like it's in a room
- ☐ Too much = sounds far away. Back it off.

4. Headroom

- ☐ Nothing hits the red on the meters
 - ☐ If something does, turn that track down (not the whole song up)
-

Level 2 mix moves (Song 2)

- ☐ **Muddy?** Turn down the bass/low end on everything except kick and bass
 - ☐ **Dull vocal?** Add a little treble
 - ☐ **Some words disappear?** Use a vocal preset with compression, or add a little more
 - ☐ **Low-volume test:** still balanced when very quiet? Then it's a good mix
-

Check on more than one speaker

- ☐ Headphones
- ☐ iPad or phone speaker
- ☐ A car or a regular speaker, if you can

If the vocal disappears on the small speaker, turn it up.

Ending

- ☐ Ends on a big final chord, **or**

- ☐ Fades out over the last 2–4 bars (volume automation on the master track)
-

Export

GarageBand on iPad

1. Go to **My Songs** (the song browser).
2. Press and hold your song → **Share**.
3. Choose **Song**, pick **Highest Quality**, add the title and your name as artist.
4. Save to Files or send it.

GarageBand on Mac

1. Menu: **Share** → **Export Song to Disk**.
2. Choose **AAC** or **MP3, Highest Quality**.
3. Name it `S1-[title]-FINAL` .

- ☐ Exported file saved
- ☐ GarageBand project saved separately (so you can remix later)
- ☐ Lead sheet made: melody PDF from おんぷノート, with chord names, lyrics, title, and your name added in Files → Markup

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Songwriter Journal

Songwriter: _____

Fill in one entry at the end of each session. Two minutes. Keep it short.

Session entry (copy this for each session)

Session #: ____ **Date:** _____ **Song:** S1 / S2

Today I made:

A decision I made (and why):

File name I saved:

Something I learned:

Next time I want to:

Song 1 reflection (Session 6)

Title: _____

- The part I'm most proud of:

- The hardest part was:
- If I remixed this song, I'd change:
- Next song, I want to try:

Feedback — two stars and a wish

- Star:
 - Star:
 - Wish:
-

Song 2 planning (Session 7)

- **Mood (1–2 words):**
 - **Topic or title idea:**
 - **Tempo (different from Song 1):**
 - **Key (major or minor):**
 - **One thing I want to do better than Song 1:**
-

Liner notes (Session 12)

Write 3–5 sentences for each song: what it's about, one decision you're proud of, one problem you solved.

Song 1:

Song 2:

Songwriter portfolio (Session 12)

Songs finished: _____

Skills I have now (check them):

- ☐ Map a song's sections by ear
- ☐ Choose and play a chord progression
- ☐ Build a drum groove
- ☐ Write a hook that passes the One-Listen Test
- ☐ Write lyrics that fit a melody
- ☐ Write a verse that contrasts with the chorus
- ☐ Arrange a full song with an energy curve
- ☐ Record and mix a song
- ☐ Use a minor key or borrowed chord on purpose
- ☐ Write a bassline
- ☐ Use syncopation
- ☐ Write a pre-chorus and a bridge
- ☐ Use production tricks (drops, risers, automation)

My signature move:

Next I want to learn:

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Listening List

Songs to study, grouped by the idea they show well. You don't need all of them. Pick one or two per session.

Always better: use songs he already loves. Any song with a clear verse and chorus works for mapping — J-pop, anime openings, game music with vocals, K-pop. This list is a backup.

Producer note: preview for content before playing. Use clean/radio edits where they exist.

Song structure and hooks (Sessions 1, 3)

- "Happy" — Pharrell Williams
- "Shake It Off" — Taylor Swift
- "Firework" — Katy Perry
- "Counting Stars" — OneRepublic

The four-chord loop (Session 1)

- "Let It Be" — The Beatles (I-V-vi-IV)
- "Someone Like You" — Adele (I-V-vi-IV)
- "Don't Stop Believin'" — Journey (I-V-vi-IV in the verse)
- "Despacito" — Luis Fonsi ft. Daddy Yankee (vi-IV-I-V)
- "Stand By Me" — Ben E. King (I-vi-IV-V)
- "Viva la Vida" — Coldplay (IV-V-I-vi)

Grooves (Session 2)

- Basic rock beat: "Shake It Off" — Taylor Swift
- Four on the floor: "Levitating" — Dua Lipa

- Dembow: "Despacito" — Luis Fonsi ft. Daddy Yankee

Pre-chorus (Sessions 5, 9)

- "Call Me Maybe" — Carly Rae Jepsen
- "Firework" — Katy Perry

Syncopation and flow (Session 9)

- "Uptown Funk" — Mark Ronson ft. Bruno Mars
- "Shape of You" — Ed Sheeran

Minor mood (Session 7)

- "Rolling in the Deep" — Adele
- "Believer" — Imagine Dragons
- "Zombie" — The Cranberries (same four chords as vi-IV-I-V, but it feels minor — a great "which chord is home?" debate)

Borrowed chords (Session 7)

- "In My Life" — The Beatles (listen for the minor iv chord near the end of the verse)
- "Hey Jude" — The Beatles (the long "na na na" ending uses $\flat$ VII: I- $\flat$ VII-IV-I)
- The *Super Mario Bros.* level-complete fanfare (ends $\flat$ VI- $\flat$ VII-I: the "Mario cadence")

Basslines and riffs (Session 8)

- "Another One Bites the Dust" — Queen
- "Billie Jean" — Michael Jackson
- "Seven Nation Army" — The White Stripes (a riff that sounds like a bassline)

Bridges and key changes (Session 10)

- "Love Story" — Taylor Swift (bridge, then a key change up for the final chorus)
- "Livin' on a Prayer" — Bon Jovi (key change in the final chorus)
- "Man in the Mirror" — Michael Jackson (key change)

Production (Session 11)

- "Bad Guy" — Billie Eilish (drop-outs, ear candy, unusual sounds)
- "Blinding Lights" — The Weeknd (synth layers, energy curve)

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Minor Keys & Borrowed Chords

Every major key has a minor twin

They use the **same notes**. Only the "home" chord changes.

- C major ↔ **A minor**
- G major ↔ **E minor**

A song feels minor when the **minor chord is home** — where phrases start and rest.

Same chords, different home:

- C – G – Am – F → home is C → sounds **major**
- Am – F – C – G → home is Am → sounds **minor**

Minor loops

In minor, the numbers change to match the new home. Try each and write a mood word.

1. **i – VI – III – VII** — Am F C G / Em C G D — Mood: _____
2. **i – VII – VI – VII** — Am G F G / Em D C D — Mood: _____
3. **i – iv – VI – V** — Am Dm F E / Em Am C B — Mood: _____

(In loop 3, the **V** is major — E instead of Em, B instead of Bm. It pulls hard back to home.)

Minor doesn't just mean sad. It can be mysterious, angry, determined, epic, or hype.

Borrowed chords

A song in major can borrow **one** chord from the minor key for a flash of color.

Chord	In C	In G	Feeling	Try this
iv (minor four)	Fm	Cm	Bittersweet, nostalgic	C – F – Fm – C
♭VII (flat seven)	B♭	F	Big, rock, triumphant	C – B♭ – F – C
♭VI (flat six)	A♭	E♭	Cinematic, heroic	C – A♭ – B♭ – C

The last one (♭VI – ♭VII – I) is the **Mario cadence** — it ends the *Super Mario Bros.* level-complete fanfare.

Rule of thumb: one borrowed chord is color. Three starts to sound like a different key.

Playing them

Letter names (for the chord chart and GarageBand keyboard)

- Fm = F A♭ C
- B♭ = B♭ D F
- A♭ = A♭ C E♭
- E (major) = E G# B

Ukulele (G C E A strings, top to bottom)

- Fm = 1013
 - B♭ = 3211
 - A♭ = 1343
 - E = 1402, or the easier E7 = 1202
 - Key of G versions: Cm = 0333 · F = 2010 · E♭ = 0331 · B7 = 2322
 - Minor loops in A minor: Am 2000 · F 2010 · C 0003 · G 0232 · Dm 2210
-

Color chords (bonus)

Add a seventh to any chord for a smoother, "lo-fi" or R&B sound (ukulele shapes in brackets):

- Cmaj7 = C E G **B** [0002]

- Fmaj7 = F A C **E** [2410]
 - Am7 = A C E **G** [0000]
 - Dm7 = D F A **C** [2213]
-

My Song 2 progression: _____ - _____ - _____ - _____

Mood I'm going for: _____

Home chord: _____ (major / minor)

Song Lab

Hear chords, build beats, and map your song. Scan the code or type:

cmfnyc.github.io/song-lab/



Bass & Counter-Melody Builder

Bass: four levels

Go up one level at a time. Stop when it sounds good. Simple often wins.

Level 1 — Roots

Play the chord's letter name on beat 1. (C chord → play C.)

Level 2 — Lock with the kick

Play the root, but copy the **kick drum's rhythm**. When bass and kick hit together, the groove feels tight.

Level 3 — Root + fifth or octave

Bounce between the root and the note five steps up (C → G), or the same note higher (low C → high C).

Level 4 — Walk to the next chord

On the last beat of the bar, step toward the next chord's root. Example, C going to G: C ... E – F – **G**.

Example (key of C, progression C – G – Am – F)

- **Level 1:** C | G | A | F — one long note per bar
 - **Level 2:** C C . C (matches a Basic Rock kick)
 - **Level 3:** C – G – C – G | G – D – G – D | A – E – A – E | F – C – F – C
-

Bass tips

- **Simpler under singing.** Busy bass is fine in the intro and between lines.
 - **Low-note mud:** if chords and bass are both low, move the chords up an octave.
 - **Pedal tone** (Level 2 trick): hold the home note in the bass while chords change on top. Creates tension.
-

Counter-melody: two rules

A counter-melody is a second melody that fills the gaps in the main one.

1. **When the lead moves, the counter-melody holds or rests.**
2. **When the lead holds or rests, the counter-melody moves.**

Put it **higher or lower** than the lead and use a **different sound** (bells, pluck synth, guitar, strings).

A short counter-melody that repeats is a **riff**. Riffs make great intros.

Where could my counter-melody go?

- ☐ Intro riff
 - ☐ In the gaps between chorus lines
 - ☐ After each verse line
 - ☐ Answering the vocal hook (call and response)
 - ☐ Bridge or instrumental section
-

In おんぷノート

The app keeps one score at a time, so each part gets its own score and its own PDF.

1. Save your current score as a PDF first.
2. Start a new score with a **bass clef**. Pick a bass-like playback sound.

3. Check each bass note against your chord chart. The note on beat 1 should usually be the chord's root.
4. Export the PDF (S2-[title]-bass-v1.pdf).
5. For the counter-melody: new score, treble clef, bright sound, export the PDF.
6. Hear everything together by playing each part into GarageBand, reading from your PDFs.

My bass level: 1 / 2 / 3 / 4 **My counter-melody sound:** _____

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Bridge, Lift & Production Tricks

The bridge: something different

By the bridge, the listener has heard the verse and chorus twice. Give them something new, then send them back to the chorus fresh.

Pick **two or three** kinds of contrast:

- ☐ **New chords** — start on a chord the song hasn't started on (IV or vi is common)
- ☐ **New melody shape** — lower and wordier, or higher and longer, than the chorus
- ☐ **New lyric angle** — a twist, a realization, the other side of the story
- ☐ **New energy** — drop way down, or build way up

End the bridge on V (G in C, D in G). It pulls strongly back to the chorus.

Bridge chord ideas (key of C):

- F – G – Am – Am (IV – V – vi – vi)
- Am – F – C – G (vi – IV – I – V)
- F – Fm – C – G (borrowed iv)

What hasn't my song said yet? _____

The lift: make the last chorus the biggest

- ☐ **Key change** — move the final chorus up 1 or 2 semitones
- ☐ **New layer** — harmony vocals, octave-up melody, a riff, bigger drums
- ☐ **Drop and return** — first half of the chorus is just vocal + one instrument, then everything comes back
- ☐ **Ad-libs** — the singer improvises around the melody

No bridge? A **double final chorus** with a lift works too.

Production tricks

Arrangement is subtraction. Something should change every 4 or 8 bars.

Pick **at least three**:

- ☐ **Drop-out** — mute everything except the vocal (or drums) for 1–2 bars, then bring it back
 - ☐ **Build / riser** — snare roll, rising synth, or reverse cymbal before a chorus. Search Apple Loops for "riser," "sweep," or "uplifter"
 - ☐ **Filter sweep** — chords sound muffled, then open up
 - ☐ **Volume automation** — a pad fades in over the pre-chorus; the chorus gets a little louder
 - ☐ **Ear candy** — a tiny one-time sound: a "hey," a reversed note, a sound effect
 - ☐ **Sampling** — record a household sound in GarageBand's Sampler and play it as an instrument
 - ☐ **Octave double** — play the hook or riff an octave higher on a new sound in the chorus
 - ☐ **Air pocket** — one beat of silence right before the chorus
-

Transitions

At every section change, give a signal:

- A drum fill
 - A crash cymbal on beat 1 of the new section
 - A riser
 - A drop-out
 - A one-beat stop
-

Reference check

Play your song, then a song you admire. Compare:

- Is my vocal as clear?
- Is my chorus as big?
- Does something change as often?

One thing I'll fix: _____

Song Lab

Hear chords, build beats, and map your song. Scan the code or type:

cmfnyc.github.io/song-lab/



Using おんぷノート (Onpu Note) for Songwriting

おんぷノート is Kawai's free notation app for school iPads. You write notes with a finger or stylus and the app turns them into clean notation. Then it plays them back.

It's great for **writing a melody down and hearing it**. It wasn't built for songwriting, so a few jobs happen outside the app. This page shows how.

What it does well

- **Handwriting input:** write notes on the staff and the app converts them
- **Instant playback**, even before a bar is finished
- **34 instrument sounds** (try piano, flute, strings, or a synth-like sound for a pop lead)
- **Tempo slider** for playback speed
- **Treble and bass clef**, key signatures, time signatures, ties, dotted notes, rests
- **PDF export** to Files, iCloud Drive, or Google Drive

What it can't do (and what to do instead)

It can't...	Do this instead
Save more than one score	Export a PDF at the end of every session, before starting anything new
Show chord names	Keep a chord chart on paper with bar numbers that match the score
Hold lyrics	Write lyrics on paper with one dash per note (see Lyric Toolkit)
Export MIDI to GarageBand	Play the melody in on a GarageBand keyboard, reading your score
Print	Export the PDF and print it from Files

Rule 1: One score at a time → PDF every session

The app only keeps one score. Starting a new one can replace the old one.

At the end of **every** session:

1. Export the score as a **PDF**.
2. Name it with the course pattern: `S1-[title]-chorus-melody-v1.pdf`
3. Save it to a folder **you** control (iCloud Drive or Google Drive). School iPads can be reset or managed by the school, so keep your own copy.

Before writing a bassline or starting Song 2, check that the current melody PDF is saved. Then clear the score and start the new part.

Rule 2: Chords live on a chord chart

Since the app has no chord names, keep a simple chart beside the iPad. Number the bars to match the score:

Bar:	1	2	3	4	5	6	7	8
Chord:	C	G	Am	F	C	G	Am	F
Notes:	CEG	GBD	ACE	FAC	CEG	GBD	ACE	FAC

The **Notes** row is the key part. It lets you check the melody by eye: **the note on beat 1 of each bar should usually be one of that bar's chord notes**.

Rule 3: Check the melody against the chords by ear

The app plays the melody but not the chords. Two ways to hear them together:

- **Producer plays chords:** set the playback tempo slow. You (or he) play the chord chart on the instrument while the app plays the melody.

- **Play it into GarageBand (best):** he plays the melody on a GarageBand keyboard track over his groove, reading from the score. It's great sight-reading practice, and now the melody is in the song.
-

Rule 4: Getting the melody into GarageBand

There's no MIDI export, so:

1. **Play it in (recommended).** In GarageBand, add a Keyboard track, pick a lead sound, turn on the metronome, and play the melody from the score. Slow the song tempo down to record if needed, then speed it back up. GarageBand keeps it in time. On the Mac, fix wrong notes afterward in the piano roll.
 2. **Record the playback (backup).** If おんぷノート is on the school iPad and GarageBand is on a different device, hold the iPad near the other device's mic and record the playback. The app's tempo slider isn't set in exact BPM, so this can drift out of time with the beat. Treat it as a sketch, not a final track.
-

Rule 5: The lead sheet is a marked-up PDF

A lead sheet shows melody, chord names above the staff, and lyrics below. To make one:

1. Export the finished melody as a PDF.
 2. Open it in the **Files** app and tap the **Markup** (pen) button.
 3. Handwrite the **chord names** above the staff at the start of each bar.
 4. Handwrite the **lyrics** under the notes, one syllable per note.
 5. Add the **title** and "**Music and lyrics by [name]**" at the top.
 6. Tap **Done** and save it as `S1-[title]-LEAD-SHEET.pdf`.
-

Handwriting tips

- **Zoom in** before writing so noteheads land clearly on a line or space.
- A **stylus** (Apple Pencil or any iPad stylus) is much more accurate than a finger.
- Write **one bar, then play it back**. Catch recognition mistakes early.

- If a note is recognized wrong, erase it and write it again, a little bigger.
 - Set the **key signature** and **time signature (4/4)** first. Then you'll need fewer sharps and flats.
-

Setting the tempo

The playback speed is a slider, not a BPM number. Match it by ear: play the GarageBand metronome at the song's BPM, then move the slider until the app's playback feels the same speed.

If he outgrows it

If he wants chord symbols, lyrics, many saved scores, or MIDI export, these free or freemium options are worth a look. Check that they suit the device first.

- **GarageBand on Mac:** its score view shows notation for any keyboard part he plays in.
- **Flat** (flat.io) or **Noteflight** (noteflight.com): browser-based notation with chord symbols and lyrics. Their free tiers limit how many scores you can keep.

Song Lab

Hear chords, build beats, and map your song. Scan the code or type:

cmfnyc.github.io/song-lab/



Ukulele Songwriter's Guide

The ukulele is a great songwriting instrument. The most famous pop progression, **C - G - Am - F**, uses four of the easiest chords on it.

Your home key: C. Use the key of C for Song 1. The key of G also works well on ukulele if a song sits better for your voice there.

How to read the chord shapes

Four numbers, one for each string, in the order you hold the ukulele: **G - C - E - A** (top string to bottom string).

- **0** = open string (no finger)
- **2** = press at the 2nd fret

Example: **Am = 2000** → 2nd fret on the G string, everything else open.

Key of C chords (Song 1)

Chord	Number	Shape (G C E A)	Difficulty
C	I	0003	easy
Dm	ii	2210	easy
Em	iii	0432	medium
F	IV	2010	easy
G	V	0232	easy
Am	vi	2000	easy

The hit-song loop: C (0003) → G (0232) → Am (2000) → F (2010)

Smooth-change tip: going from Am (2000) to F (2010), keep your 2nd-fret finger down and just add one finger. Look for fingers that can stay put when you change chords.

Key of G chords (optional)

Chord	Number	Shape (G C E A)
G	I	0232
Am	ii	2000
Bm	iii	4222 (barre, harder)
C	IV	0003
D	V	2220
Em	vi	0432

Phase 2 chords (Session 7)

In the key of C:

- Fm (iv, bittersweet) = **1013**
- B^b (bVII, triumphant) = **3211** (1st-fret barre across the bottom two strings)
- A^b (bVI, cinematic) = **1343**
- E (III, pulls to Am) = **1402**, or the easier E7 = **1202**

In the key of G:

- Cm (iv) = **0333**
- F (bVII) = **2010**
- E^b (bVI) = **0331**
- B7 (instead of B, pulls to Em) = **2322**

Color chords (smooth, lo-fi sound): Am7 = **0000** (just strum the open strings), Cmaj7 = **0002**, Dm7 = **2213**, Fmaj7 = **2410**

Strum patterns = chord rhythm

These match the chord rhythms in Song Lab and the Beat Patterns handout. D = down strum, U = up strum, **X** = chuck.

1. Whole notes (ballad)

Count:	1	&	2	&	3	&	4	&
Strum:	D	.	.	.	.	.	.	.

2. Quarters (driving)

Count:	1	&	2	&	3	&	4	&
Strum:	D	.	D	.	D	.	D	.

3. The push (modern pop)

Count:	1	&	2	&	3	&	4	&
Strum:	D	.	.	D	.	.	.	.

Strum on 1, then again on the "&" of 2, and let it ring.

4. Island strum (classic ukulele)

Count:	1	&	2	&	3	&	4	&
Strum:	D	.	D	U	.	U	D	U

Keep your hand moving down-up the whole time, even when you miss the strings.

5. The chuck (the ukulele's built-in snare drum)

Count:	1	&	2	&	3	&	4	&
Strum:	D	.	X	.	D	U	X	.

How to chuck: strum down and land the side of your palm on the strings right away, muting them with a percussive "chk." Putting the chuck on beats 2 and 4 is exactly where the snare goes. It makes a solo ukulele sound like a band.

Finding melody notes on the ukulele

The ukulele's top G string is tuned **high** (called "re-entrant" tuning), so the **lowest note is the open C string: middle C**. That means ukulele melodies sit right on the treble staff, perfect for おんぷノート.

Notes of the C major scale, first five frets:

- **C string:** open = C (middle C) · 2 = D · 4 = E · 5 = F
- **E string:** open = E · 1 = F · 3 = G
- **A string:** open = A · 2 = B · 3 = C · 5 = D
- **G string (high G):** open = G · 2 = A · 4 = B · 5 = C

Scale path, low to high: C string 0, 2, 4 → E string 1, 3 → A string 0, 2, 3 = **C D E F G A B C**

Hum a note, then hunt for it on this path. Write down the letter name, then put it in おんぷノート.

What the ukulele can't do: bass

Because of the high G string, the ukulele has no low notes. For basslines (Session 8):

1. Find the bass note's **letter name** on the ukulele (the chord's root).
 2. Play it on a GarageBand **Bass** or **Keyboard** track, down low.
 3. Or use GarageBand's Smart Bass and tap the chord strips.
-

Recording your ukulele in GarageBand

A real ukulele track makes the song sound like **you**.

1. **Tune first.** GarageBand has a built-in tuner, or use a free tuner app.
2. Add an **Audio Recorder** track.
3. **Headphones on**, metronome on.
4. **Mic placement:** about 15–20 cm away, pointed at where the neck meets the body (not straight into the sound hole, which sounds boomy).

5. Record 3 takes. Keep the steadiest one.

6. Mistake in one bar? Record just that section again.

Ukulele as the lead: don't want to sing? Pluck the melody on the ukulele and record it as the lead track.

Ukulele songwriting tricks

- **Capo** (if you have one): clip it on the 2nd fret and play the same C-shapes. The song moves up to the key of D, handy if the melody is too low for your voice.
- **Riff from a chord:** hold a chord shape and pluck its strings one at a time in a pattern. Instant intro riff.
- **Stop-time:** in the pre-chorus, strum once per bar and mute. Then burst into the full strum on the chorus.
- **Two ukulele parts:** record strummed chords, then a second track of plucked notes. Pan them slightly apart.